

Run Claude Code on Gemini — Setup Guide

Use Gemini models inside Claude Code through **Claude Code Router** (`ccr`), a small proxy that translates between Claude Code's format and Gemini's.

Requires **Node.js** installed first. Check with `node --version` .

Step 1 — Install the tools

```
npm install -g @anthropic-ai/claude-code @musistudio/claude-code-router
```

Confirm both installed:

```
claude --version
ccr -v
```

Step 2 — Get a free Gemini API key

1. Open <https://aistudio.google.com/>
2. Click **Get API key** (top right)
3. Create a project, then create the key
4. Copy the key somewhere safe

Step 3 — Create the config file

Create the folder and file:

```
mkdir -p ~/.claude-code-router
```

Then put this in `~/.claude-code-router/config.json` (on Windows: `C:\Users\YourName\.claude-code-router\config.json`). Paste your Gemini key where shown:

```
{
  "LOG": true,
  "HOST": "127.0.0.1",
  "PORT": 3456,
  "APIKEY": "pick-any-local-password",
```

```
"Providers": [
  {
    "name": "gemini",
    "api_base_url": "https://generativelanguage.googleapis.com/v1beta/models/",
    "api_key": "YOUR_GEMINI_API_KEY",
    "models": ["gemini-2.5-flash", "gemini-2.5-flash-lite"],
    "transformer": { "use": ["gemini"] }
  }
],
"Router": {
  "default": "gemini,gemini-2.5-flash-lite",
  "background": "gemini,gemini-2.5-flash-lite",
  "think": "gemini,gemini-2.5-flash",
  "longContext": "gemini,gemini-2.5-flash"
}
}
```

The `"transformer": { "use": ["gemini"] }` line is what does the format translation.

Step 4 — Start the router

In a terminal, run:

```
CCR start
```

Wait 2–3 seconds until it says the service has started.

Keep this terminal open. `CCR start` runs the proxy as a live service — it stays in the foreground and will look like it's "stuck," but that's normal. **Do not close this window;** closing it stops the router. Just leave it running in the background.

Step 5 — Launch Claude Code through the router

Open a **new terminal tab/window** (don't reuse the `CCR start` one), then run:

```
CCR code
```

Use `CCR code` instead of `CLAUDE` from now on. Everything else works the same.

Daily commands

Command	What it does
<code>ccr start</code>	Start the proxy
<code>ccr code</code>	Launch Claude Code through Gemini
<code>ccr restart</code>	Reload after editing <code>config.json</code>
<code>ccr ui</code>	Edit config in a web UI
<code>/model gemini,gemini-2.5-flash</code>	Switch model live inside a session

Troubleshooting

- **ccr: command not found** → npm's global bin isn't on PATH. Reopen the terminal.
- **Config change ignored** → run `ccr restart` (live `/model` works instantly, file edits don't).
- **"Rate Limit" errors** → use **Flash-Lite** as default. Claude Code fires many small background calls; low-RPM models (Flash/Pro) throttle almost immediately on the free tier.
- **Timeout on `ccr code`** → run `ccr start` separately first, then `ccr code`.
- **Port already in use** → a stale `ccr` process is holding port 3456; kill it and retry.
- **Scripted/CI runs hang** → add `"NON_INTERACTIVE_MODE": true` to the config.

Important notes

- **Use Flash-Lite for free.** Pro models are paywalled and burn the free daily quota in minutes.
- **Free tier quota is per Google project, not per key** — extra keys don't add quota. For a workshop, each person needs their own Gemini project to avoid throttling each other.
- **Enabling billing removes the free tier** on that project — every call becomes billable from the first token.
- **Claude Code Router is a community project**, not official Anthropic software. Fine for experimenting; avoid putting secrets or production code behind it without review.
- Non-Claude models can behave inconsistently in Claude Code (tool calls, formatting), since Claude Code is tuned for Claude.